

Proteomics signature of whole-brain microstructural integrity of white matter using LatentSNA

#13785

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- White matter microstructural integrity is biologically relevant, as FA-weighted connectomes may capture early Alzheimer’s disease (AD)-related axonal, myelin, glial, and vascular injury.
- Plasma proteomics provides a scalable readout of peripheral molecular processes that may be linked to brain tissue vulnerability.
- LatentSNA modeled high-dimensional SomaScan 7K proteins in relation to whole-brain FA-weighted structural connectome variation among 139 IADRC participants.
- Protein associations were region-specific, demonstrating distinct positive and negative patterns across white matter networks rather than a single global effect.
- Prominent protein–region signals included FLNA, BLVRB, and HMBS, with higher protein levels associated with lower region-specific white matter connectivity in the test set.
- Immune-cell/protein relationships converged on the frontal pole, lingual, lateral occipital, and right caudate regions, supporting immune-related proteins as candidate markers of AD-related white matter vulnerability.
- These findings motivate replication in larger longitudinal cohorts models to evaluate biomarker robustness and disease relevance.

BACKGROUND

Diffusion MRI fractional anisotropy (FA)–weighted structural connectomes quantify region-to-region variation in white matter microstructural integrity and may capture early Alzheimer’s disease–related neurodegeneration.

Plasma proteomics enables high-dimensional profiling of peripheral molecular processes potentially linked to brain tissue injury.

However, the extent to which high-dimensional circulating proteomic profiles are associated with spatially distributed variation in FA-weighted white matter structural connectomes remains poorly defined.

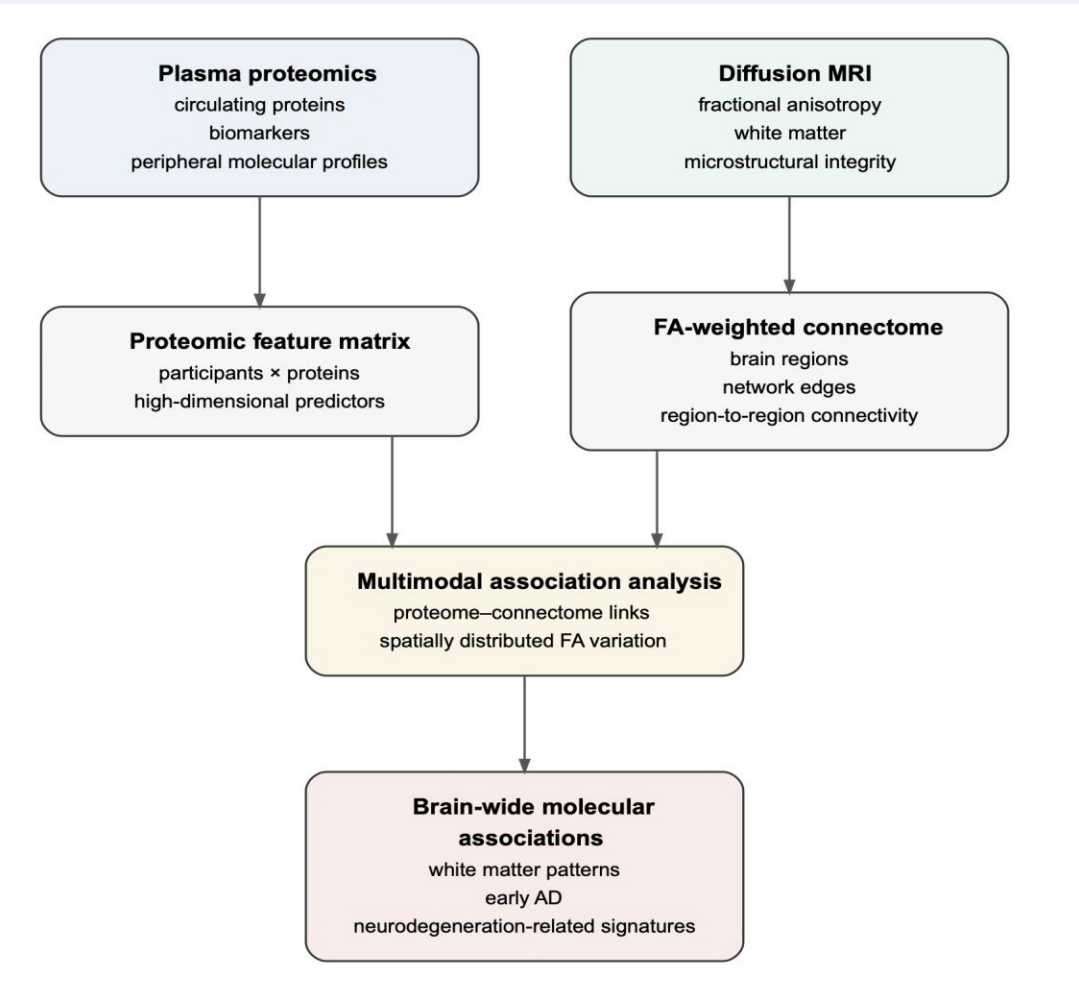


Figure: Conceptual workflow illustrating the integration of plasma proteomic profiles with FA-weighted structural connectomes.

We use LatentSNA (Wang et al., 2025) to model protein associations with latent patterns of FA-based white matter network variation.

METHODS

We analyzed 139 Indiana Alzheimer’s Disease Research Center (IADRC) participants with paired SomaScan 7K v4.1 plasma proteomics and diffusion MRI–derived white matter connectomes.

- Proteins were QC-filtered, normalized, log-transformed, and standardized.
- Structural connectomes were represented as FA-weighted symmetric adjacency matrices.
- Nodes were defined by the study parcellation.
- Edges indexed tract-level white matter microstructural integrity.

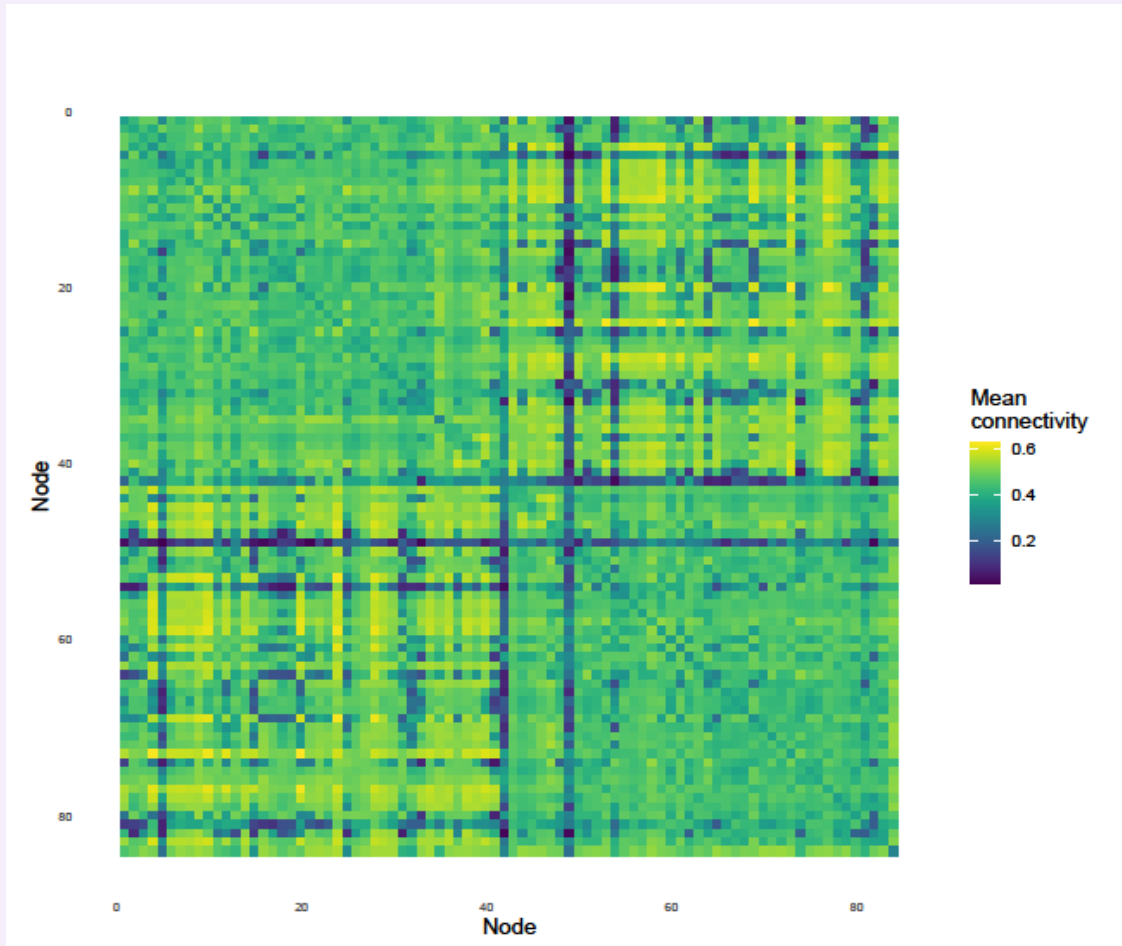


Figure: Mean Connectivity across the 139 structural connectomes of 84 different brain regions

For each protein, we fit an unadjusted LatentSNA model to estimate its marginal association with whole-brain FA network variation.

- LatentSNA is a Bayesian latent-space network model.
- It jointly decomposes observed connectomes and behavioral measures into latent signal plus noise.
- It represents brain-network topology with bilinear latent effects.
- It infers brain–behavior biomarkers from the covariance between latent connectivity variables and latent behavioral traits.

Proteins were ranked by standardized LatentSNA association strength, and protein-associated latent network patterns were mapped back to brain regions. Top-ranked proteins were mapped to protein-coding genes and

annotated using external Human Protein Atlas reference resources, including immune-cell ribonucleic acid expression data and normal-tissue immunohistochemistry protein-expression data.

- The immune-cell expression data provided transcriptomic context across curated immune-cell populations.
- The immunohistochemistry data provided normal-tissue and annotated cell-type protein-expression evidence.
- The immunohistochemistry data included brain-region annotations where available.

Annotation results were used only for biological interpretation and visualization, not as inputs to the LatentSNA model. LatentSNA association strength.

RESULTS

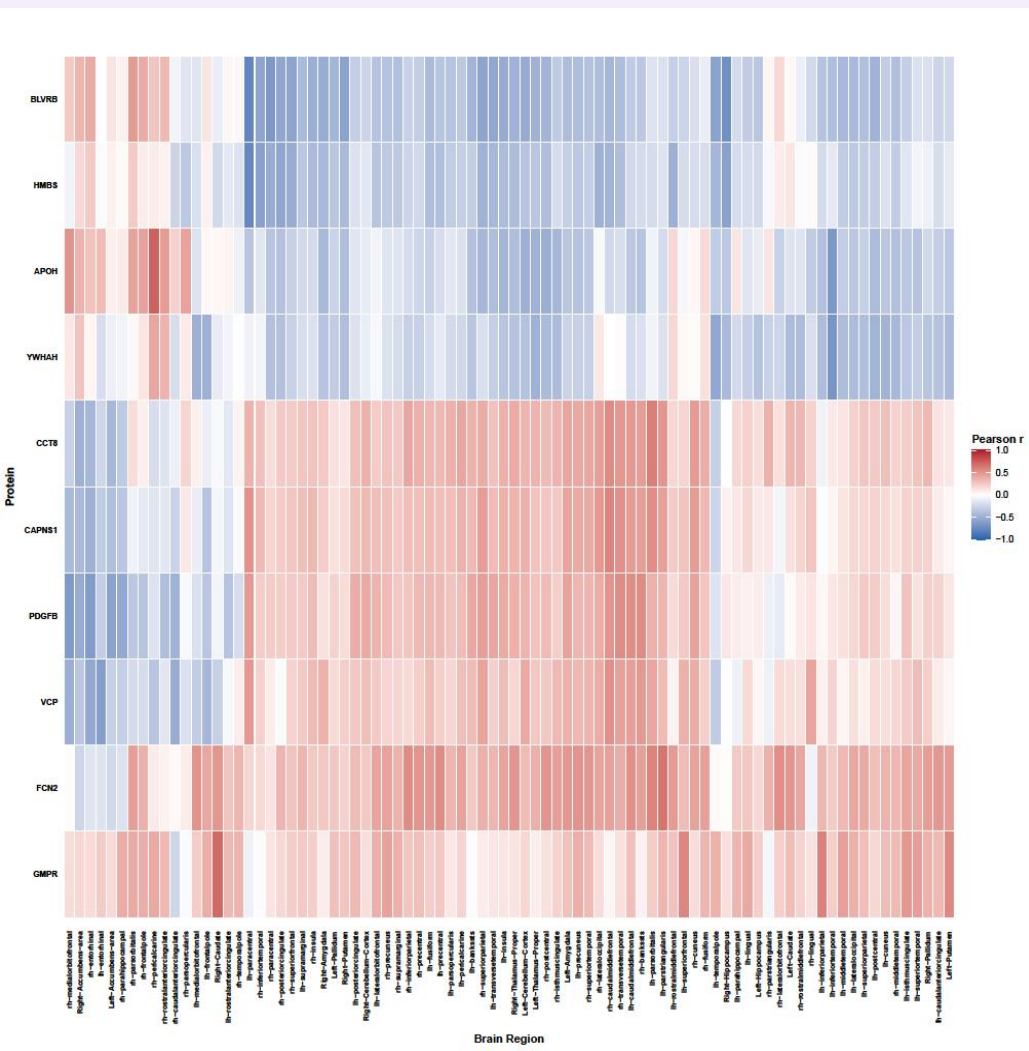


Figure: Heatmap of the correlation matrix between observed plasma proteins and model-derived regional brain scores of the test participants. The plot was restricted to the highest 10 annotated proteins for readability. Red indicates positive correlation and blue indicates negative.

Plasma protein associations varied by brain region, showing distinct positive and negative patterns across white matter networks.

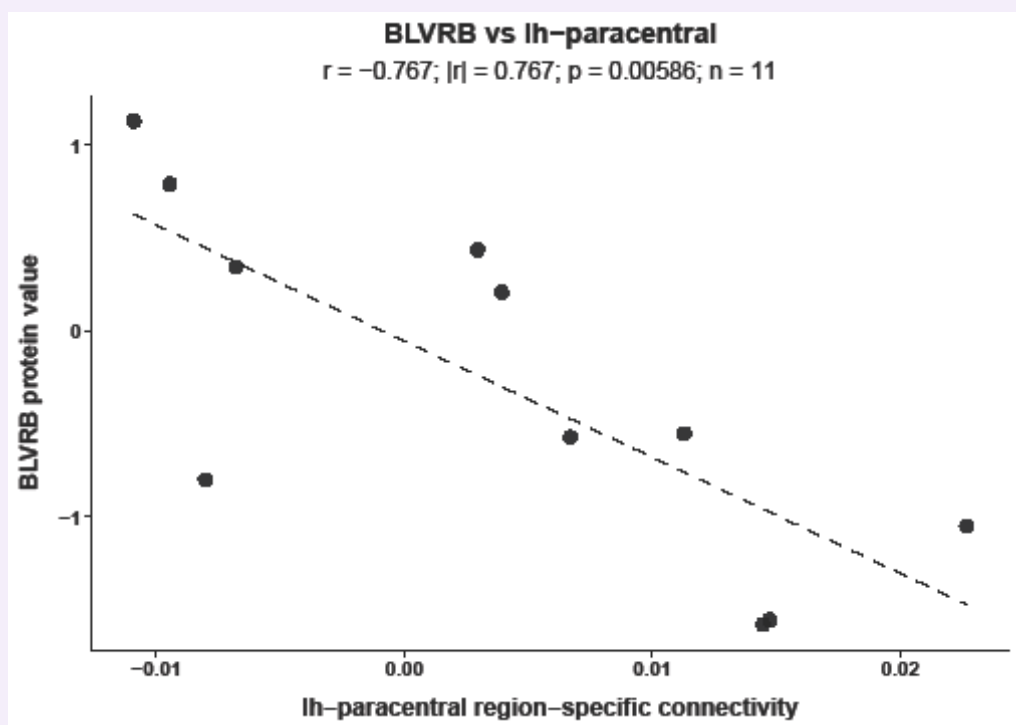
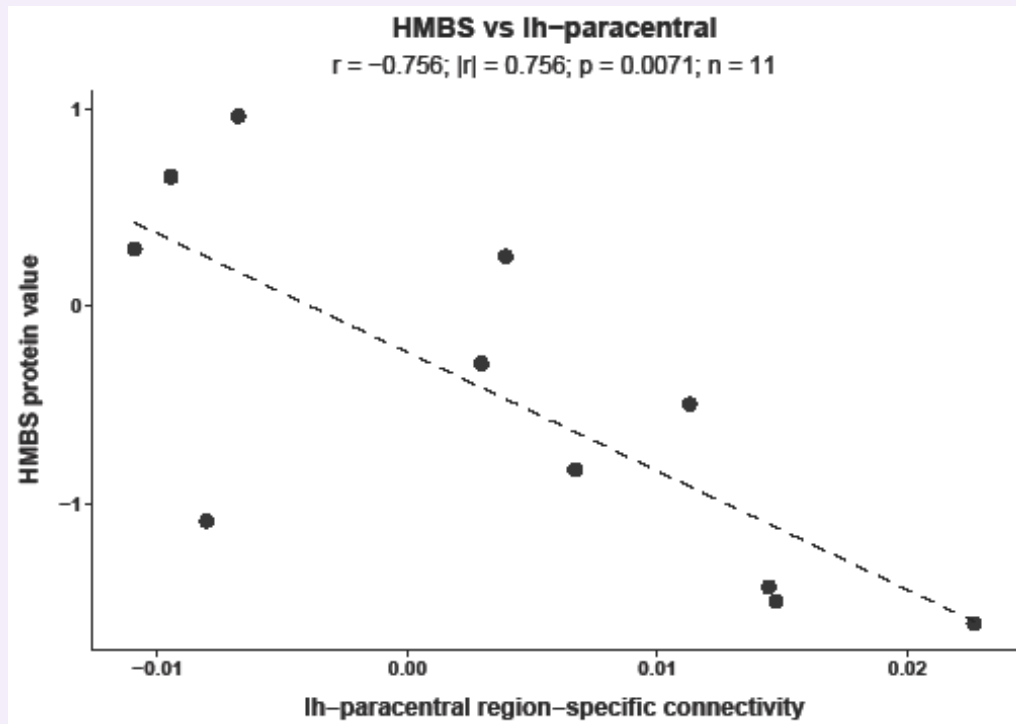
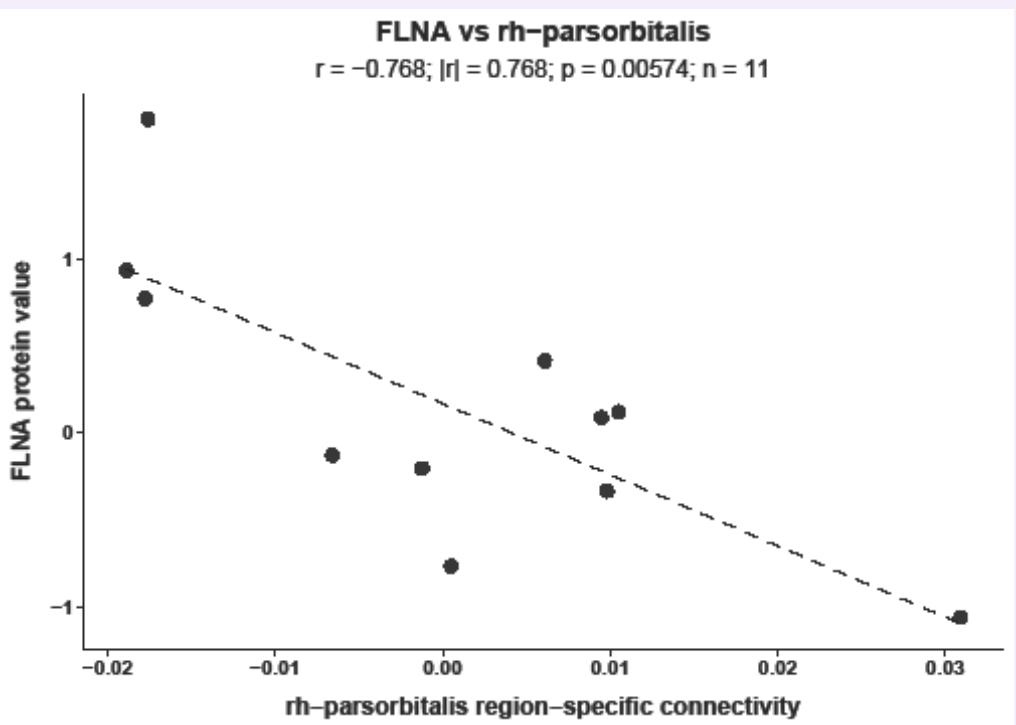


Figure: Scatterplots of the three highest protein–brain region associations ranked by absolute correlation of latent brain and observed protein scores among test participants from the selected model.

Higher FLNA levels were strongly associated with lower white matter connectivity in the right pars orbitalis, whereas higher BLVRB and HMBS levels were strongly associated with lower connectivity in the left paracentral region in the test set.

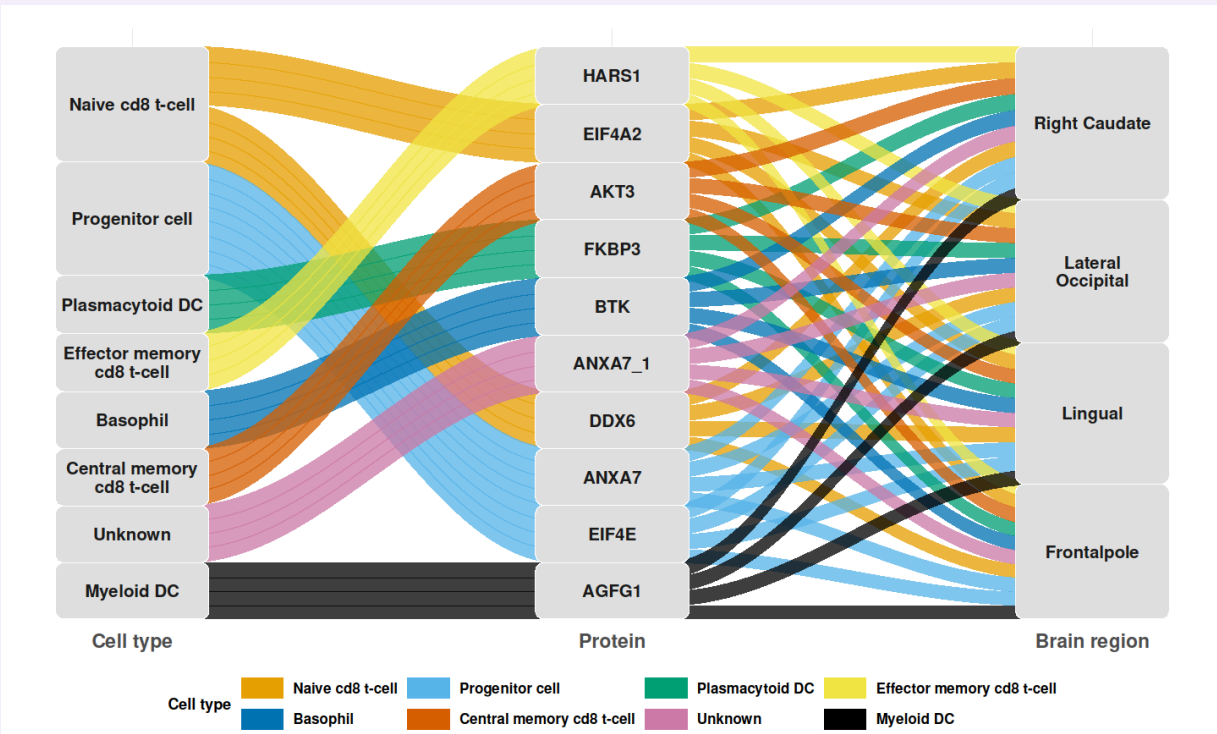


Figure: Alluvial plot of the strongest immune-cell, protein, and brain-region relationships from the selected model, limited to the top 10 proteins by latent–observed brain-score correlation. Flow weights show absolute protein–region correlations.

The selected model identified strongest immune-cell–protein relationships that converged on frontal pole, lingual, lateral occipital, and right caudate regions.

CONCLUSION

LatentSNA identified plasma proteomic signatures associated with region-specific white matter microstructural integrity, supporting immune-related proteins as candidate markers of Alzheimer’s disease–related white matter vulnerability.

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CONTACT